

Declared Stratification and Tolerance-Slice Holdout Replication in a Prime-Shell Diagnostic

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Abstract

We test whether a single balanced aggregation rule can replicate a finite prime-shell preference across a held-out tolerance slice. The input is the locked TPC-309 atlas of three source-backed profile ladders, three adjacent shell transitions, two kernel exponents, three tolerances, and three finite completion radii. The declared rule first pools the three profile ladders inside each fixed design cell and then gives every design cell equal arithmetic weight. At the primary native radius, the calibration slice $\{0.25, 0.5\}$ has ratio interval $[4.0615814676, 4.0617439341]$ and is left-lower, whereas the held-out slice $\{0.75\}$ has interval $[0.6818442327, 0.6818715070]$ and is right-lower. The strict class is therefore reversed. Including adversarial radii one and two leaves the calibration block left-lower but makes the confirmation block unresolved. We prove the finite two-stage interval algebra and independently replay 54 strata, six blocks, and 22 sensitivity controls. This is a scoped parameter-slice obstruction on one padded float-replay atlas, not a fresh-data, causal, asymptotic, arithmetic, or twin-prime theorem.

1 Question and route position

The recent finite diagnostics have progressively separated several sources of instability. TPC-307 introduced a common ambient operator with overlap-only fits and exclusive shell holdouts [2]. TPC-308 enumerated binary completion envelopes on those holdouts [1]. TPC-309 shifted a source-backed 17-coordinate profile window through one 19-prime pool and found that the location of strict discordance changes with the profile [4]. TPC-310 then showed that pooling losses and averaging row ratios can reverse a finite class [3].

TPC-311 asks the smallest next question suggested by that obstruction: after fixing one balanced design rule, does a preference found on two tolerance slices survive the third? The answer is deliberately scoped. The held-out slice below is a disjoint parameter slice of the same locked parent atlas; it is not an independent physical sample. Likewise, the rule is declared in this child project but is not an externally timestamped preregistration.

2 Locked data and notation

For a profile ladder $j \in \{\text{LOW}, \text{BASE}, \text{HIGH}\}$, transition $q \in \{50 \rightarrow 60, 60 \rightarrow 70, 70 \rightarrow 90\}$, exponent $e \in \{1, 2\}$, tolerance $t \in \{0.25, 0.5, 0.75\}$, and radius $r \in \{0, 1, 2\}$, the TPC-309 certificate supplies positive extrema

$$\begin{aligned} R_{j,q,e,t,r}^- &\leq R_{j,q,e,t,r}^+ \\ L_{j,q,e,t,r}^- &\leq L_{j,q,e,t,r}^+ \end{aligned}$$

They are the lower endpoint of the corresponding right/left minimum envelope and the upper endpoint of the corresponding maximum envelope. There are

$$3 \cdot 2 \cdot 3 \cdot 3 = 54$$

design cells and three profile rows in each, hence 162 locked parent observations. We do not regenerate labels, shell partitions, physical matrices, or completion balls.

The strict class map used throughout is

$$\text{class}([a, b]) = \begin{cases} \text{RIGHT}, & b < 0.9, \\ \text{LEFT}, & a > 1.1, \\ \text{UNRESOLVED}, & \text{otherwise.} \end{cases}$$

3 Declared two-stage protocol

Fix a design cell $s = (q, e, t, r)$. First pool the replicated profile ladders:

$$P_s = \left[\frac{\sum_j R_{j,s}^-}{\sum_j L_{j,s}^+}, \frac{\sum_j R_{j,s}^+}{\sum_j L_{j,s}^-} \right]. \quad (1)$$

The second stage gives each selected design cell equal weight. For a nonempty block B of cells, set

$$S(B) = \left[\frac{1}{|B|} \sum_{s \in B} P_s^-, \frac{1}{|B|} \sum_{s \in B} P_s^+ \right]. \quad (2)$$

This removes the loss-scale weights from the profile dimension, while making the design-cell weighting explicit rather than implicit.

The calibration block uses $t \in \{0.25, 0.5\}$ and the confirmation block uses $t = 0.75$. They are disjoint and together cover the declared tolerance grid. The primary endpoint is $r = 0$, the native completion. The all-radius blocks with $r \in \{0, 1, 2\}$ are adversarial stress controls, not a second primary choice. We also compute exponent slices, transition slices, leave-one- transition controls, and leave-one-ladder controls without changing the primary rule.

4 Finite statements

Lemma 1 (Independent finite extrema). *Let X_j be nonempty finite sets with extrema x_j^- and x_j^+ . If the choices are independent, then*

$$\min \sum_j x_j = \sum_j x_j^-, \quad \max \sum_j x_j = \sum_j x_j^+.$$

Proof. Termwise endpoint inequalities give both bounds. Choosing all lower, or all upper, endpoints attains the corresponding bound. $\square$

Proposition 1 (Soundness of the two-stage map). *Equations (1) and (2) are ordered positive interval enclosures for the declared finite operations.*

Proof. All four endpoint sums in (1) are positive. The lemma gives the independent numerator extrema, and quotient monotonicity reverses the denominator endpoint, proving the first enclosure. Each P_s^- is at most P_s^+ ; summing these inequalities and dividing by the positive integer $|B|$ proves the second enclosure. The two tolerance sets are visibly disjoint and exhaustive on the three-point grid. $\square$

The finite class relation between calibration C and confirmation H is called *strictly replicated* when their strict classes agree, *strictly reversed* when one is RIGHT and the other LEFT, and *unresolved-slice* otherwise. This terminology is a finite decision diagnostic, not a statistical generalization claim.

5 Primary results

The producer and an independent parser/replay agree on all 54 profile-pooled strata, six declared blocks, and 22 sensitivity blocks. Table 1 gives the primary and stress results. The native calibration interval lies entirely above 1.1, while the native confirmation interval lies entirely below 0.9. Their strict classes are consequently opposite by a large margin.

Thus the primary native relation is

$$\text{LEFT}_{\{0.25, 0.5\}} \longrightarrow \text{RIGHT}_{\{0.75\}}, \quad \text{a strict finite reversal.}$$

The all-radius stress relation is not a reversal because the confirmation interval widens across the thresholds; it is a failure of strict replication with an unresolved held-out slice.

Table 1: Declared stratified blocks.

Block	Strata	Ratio interval	Class
Calibration, native $r = 0$	12	[4.06158147, 4.06174393]	Left
Confirmation, native $r = 0$	6	[0.68184423, 0.68187151]	Right
Full, native $r = 0$	18	[2.93500239, 2.93511979]	Left
Calibration, all radii	36	[2.82703204, 6.58728887]	Left
Confirmation, all radii	18	[0.38404969, 2.90381633]	Unresolved
Full, all radii	54	[2.01270459, 5.35946469]	Left

Table 2: Native exponent controls.

Exponent	Slice	Ratio interval	Class
1	calibration	[7.94756945, 7.94788736]	Left
1	confirmation	[1.06436231, 1.06440489]	Unresolved
2	calibration	[0.17559348, 0.17560051]	Right
2	confirmation	[0.29932615, 0.29933813]	Right

6 Adversarial localization controls

The reversal is not evidence that the equal-stratum rule has become canonical. The controls locate several remaining dependencies. At native radius, the exponent slices in Table 2 have opposite calibration classes; the exponent-one confirmation is unresolved, while exponent two remains right-lower.

The native calibration class is also not profile-subset invariant. Omitting LOW, BASE, or HIGH gives respectively Left, Right, and Left; each corresponding confirmation control is Right. In particular, deleting BASE changes the calibration class from the primary Left to Right without altering the confirmation slice. Transition controls show the same localization: the calibration classes for $50 \rightarrow 60$, $60 \rightarrow 70$, and $70 \rightarrow 90$ are respectively Right, Left, and Right, whereas their confirmation classes are respectively Right, Unresolved, and Right. These are controls on the finite atlas, not multiple-testing-adjusted statistical estimates.

7 Interpretation and obstruction

The declared rule does accomplish one useful separation. Within a design cell, profile ladders are treated symmetrically; between cells, the factorial design is explicit. The exact proof is therefore reusable whenever a future source supplies positive finite extrema with the same independence structure.

It does not solve the problem exposed by TPC-310. The primary orientation does not replicate from the two-point calibration tolerance slice to the held-out third tolerance. Adding completion adversaries makes the held-out interval unresolved, and the exponent and profile controls show that the finite orientation is still structured by other axes. A weighting convention can be made explicit without becoming arithmetically justified.

There are two important scope boundaries. First, the confirmation rows come from the same locked parent certificate, so this is a parameter-slice holdout, not fresh physical data. Second, the child protocol was declared for this release but was not externally timestamped before the broader research session. The result should therefore be read as a reproducible obstruction to one fixed rule, not as an unbiased preregistered statistical test.

8 Claim firewall and conclusion

The exact claims are the 54-cell census, the independent profile-pooling extrema, the positive equal-stratum interval map, and the finite tau partition. The numerical claims are a replay of padded float-derived decimal intervals. They are not directed-rounding guarantees for the unrounded physical source. TPC-302 target-generation leakage remains inherited. No arithmetic L^2 estimate, fixed-power credit, uniform asymptotic budget, causal identification, full Gate-B passage, or twin-prime conclusion is obtained.

The scoped conclusion is:

For the locked TPC-309 finite atlas, the declared profile-pooled, design-equal stratification rule is strictly LEFT on the calibration tolerance slice but strictly RIGHT on the native held-out slice $\tau = 0.75$. It does not provide a replication-stable global preference. Closing this obstruction requires an externally justified weighting law and a genuinely fresh physical holdout.

Reproducibility. The canonical certificate is `results/tpc311_certificate.json`. The producer, independent replay, exact rational stress suite, theorem ledger, claim firewall, and Bridge-B checker are included in the project directory. The Session-named evaluator files were absent from the checkout, so no official Route-A or Route-B pass is asserted.

References

- [1] Liang Wang. Adversarial exclusive-completion envelopes for a prime-shell holdout, 2026. TPC-308 project release, Huazhong University of Science and Technology.
- [2] Liang Wang. Common-ambient union-shell holdout in a prime-shell diagnostic, 2026. TPC-307 project release, Huazhong University of Science and Technology.
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